

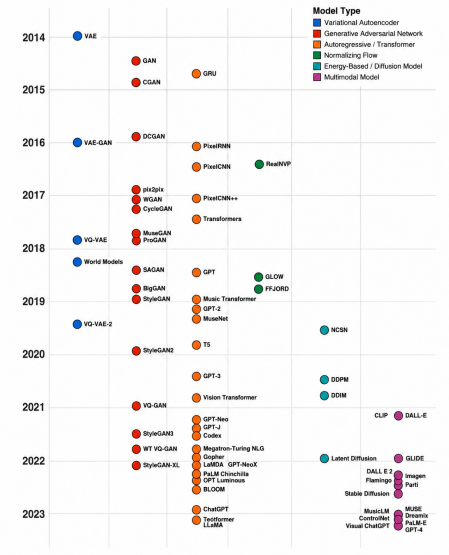


Figure: A brief history of generative AI from 2014 to 2023 (note: some important developments such as LSTMs and early energy-based models [e.g., Boltzmann machines] precede this timeline)

Write an exciting and impactful opening paragraph for a textbook about generative AI that ends on a cliffhanger.

Generative AI is one of the most revolutionary technologies of our time, transforming the way we interact with machines. Its potential to revolutionize the way we live, work, and play has been the subject of countless conversations, debates, and predictions. But what if there was an even greater potential to this powerful technology? What if the possibilities of generative AI extend beyond our current imagination? The future of generative AI may be more exciting than we ever thought possible.

Write a short one stanza rhyming poem about the following paragraph:

Until recently, discriminative modeling has been the driving force behind most progress in machine learning methodology. This is because for any discriminative problem, the corresponding generative modeling problem is typically much more difficult to tackle. For example it is much easier to train a model to predict if a painting is by Van Gogh than it is to train a model to generate a Van Gogh styled painting from scratch. Similarly, it is much easier to train a model to predict if a page of text was written by Charles Dickens than it is build a model to generate a set of paragraphs in the style of Dickens. Until recently, most generative challenges were simply out of reach and many doubted that they could ever be solved. Creativity was considered a purely human capability that couldn't be rivaled by AI.

Discriminative tasks so easily done,
Generative ones so hard to be won,
Creativity thought to be a human thing,
Until AI showed what it can bring!

Figure: Output from GPT-3—the non-highlighted text is the prompt and the green highlighted text is the output from GPT-3

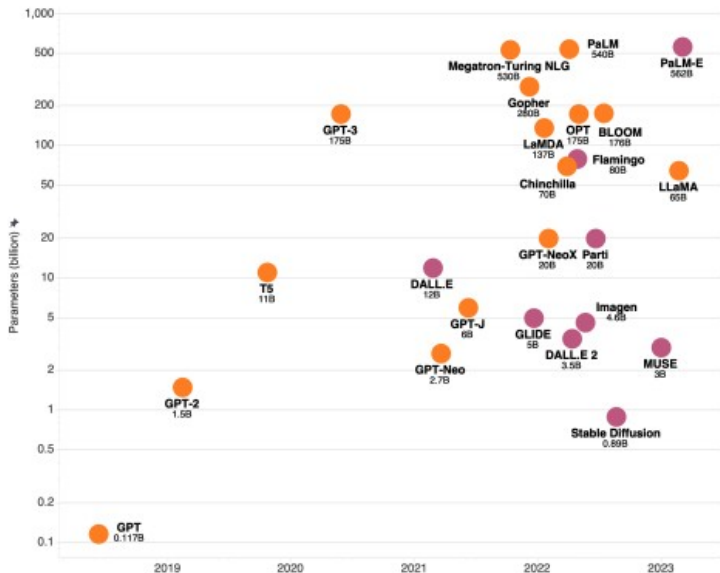


Figure: The size of large language models (orange) and multimodal models (pink) in number of parameters over time

What is the fastest time to cross the Atlantic Ocean by foot?

The fastest time to cross the Atlantic Ocean by foot is 42 days, 5 hours, and 12 minutes, set by Andrew Towne in 2020.

If I was double my sister's age when I was 8, how old am I now my sister is 70?

You are now $70 \times 2 = 140$ years old.

Figure: While large language models excel at some tasks, they are also prone to mistakes related to factual or logical reasoning (GPT-3 output shown)

```

1 import tweepy, os # secrets in environment variables
2
3 def fetch_tweets_from_user(user_name):
4     # authentication
5     auth = tweepy.OAuthHandler(os.environ['TWITTER_KEY'], os.environ['TWITTER_SECRET'])
6     auth.set_access_token(os.environ['TWITTER_TOKEN'], os.environ['TWITTER_TOKEN_SECRET'])
7     api = tweepy.API(auth)
8
9     # fetch tweets
10    tweets = api.user_timeline(screen_name=user, count=200, include_rts=False)
11    return tweets

```

Copilot

```

1 import datetime
2
3 def parse_expenses(expenses_string):
4     """Parse the list of expenses and return the list of triples (date, value, currency).
5     Ignore lines starting with #.
6     Parse the date using datetime.
7     Example expenses_string:
8         2016-01-02 -34.01 USD
9         2016-01-03 2.59 DKK
10        2016-01-03 -2.72 EUR
11    """
12    expenses = []
13    for line in expenses_string.splitlines():
14        if line.startswith("#"):
15            continue
16        date, value, currency = line.split(" ")
17        expenses.append((datetime.datetime.strptime(date, "%Y-%m-%d"),
18                        float(value),
19                        currency))
20    return expenses

```

Copilot

Figure: Two examples of GitHub Copilot capabilities (source: GitHub Copilot)

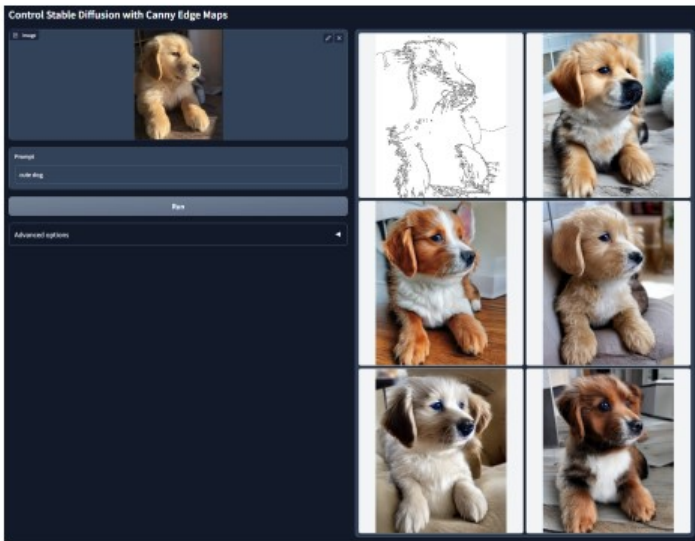


Figure: Conditioning the output of Stable Diffusion using a Canny edge map and ControlNet

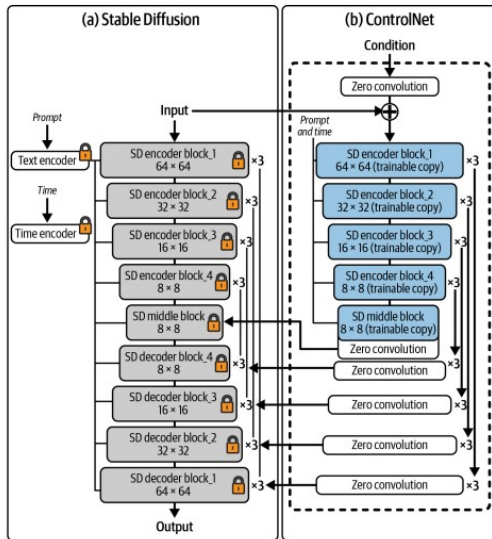
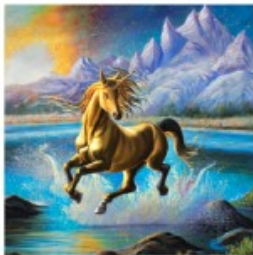


Figure: The ControlNet architecture, with the trainable copies of the Stable Diffusion encoder blocks highlighted in blue

A unicorn leaping over a magnificent river, beautiful oil painting, stunning natural scenery, award-winning art, golden lighting, intricate detail



Stable Diffusion v2.1



Midjourney



DALL.E 2

Figure: Outputs from Stable Diffusion v2.1, Midjourney, and DALL.E 2 for the same prompt

Provide me with three holiday destination suggestions in Europe that are off-the-beaten track and have both adventurous and relaxing activities nearby, for a family with teenage children. Give examples of why each is a perfect destination.

1. Riga, Latvia: Riga is a great holiday destination for a family with teenage children. It's off-the-beaten track and has something for everyone. It offers a wide range of activities, from adventure sports like kite surfing, kayaking, and rock climbing, to more relaxed activities like sightseeing and touring the city's historic sites. It's also a great base for exploring the Latvian countryside, with plenty of outdoor activities like hiking and camping.

2. Plitvice Lakes National Park, Croatia: Plitvice Lakes National Park is an ideal off-the-beaten-track destination for a family with teenage children. It offers a unique combination of activities, from relaxing boat rides and lake cruises to thrilling zip-lining and hiking. The park is also home to some of the most stunning waterfalls and cascades in Europe, making it a great place to explore and take in the incredible views.

3. Lofoten Islands, Norway: The Lofoten Islands in Norway are a great off-the-beaten-track destination for a family with teenage children. With its beautiful fjords, stunning coastal views, and abundant wildlife, there's plenty to explore in this remote archipelago. There are also plenty of activities to do, from fishing and kayaking to hiking and beachcombing. It's a great combination of adventure and relaxation.

Figure: Output from GPT-3, giving bespoke holiday recommendations

Explain the the difference between generative and discriminative AI to a high school student.

Generative AI is the process of using algorithms to generate new data based on existing data. It is primarily used for creating new ideas or objects such as music, text, images, etc. It tries to capture the underlying structure and patterns in the data and use it to generate new data from the same distribution.

Discriminative AI is the process of using algorithms to classify or distinguish between different items. It is primarily used for tasks such as image recognition, natural language processing and speech recognition. It tries to identify the differences between different items and then use those differences to classify them.

Figure: Output from GPT-3—an example of how large language models can be used for learning

The New England Journal of Medicine is a registered trademark of [QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Figure: An example of how Toolformer is able to autonomously call different APIs in order to obtain precise information where necessary